Interactive Image Style Transfer and Local Appearance Editing System

by

Zhengda SI (2230024178)

Kai WEN (2230034054)

Yiyi LIN (2230034031)

Xinyu ZHANG (2230034061)

Langqi ZHONG (2230034066)

A group project (Computer Technology and AI Project GCAP3123)

Bachelor of Science (Honours)

in

Artificial Intelligence

at

Beijing-Normal Hong Kong Baptist University

May, 2025

DECLARATION

We hereby declare that all the work done in this Project is of our independent effort. We also certify that we have never submitted the idea and product of this Project for academic or employment credits.
The contribution of each teammate is listed below:

Zhengda SI is responsible for the neural style transfer module of VGG-19, which involves feature extraction, style representation, loss function of style/content, and iterative optimization to generate stylized images without compromising the structure of the content.
Kai WEN is responsible for the stable implementation of the diffusion refinement module, which includes Img2Img generation, local refinement based on semantic segmentation, prompt control, and diffusion inference configuration.
Yiyi LIN is responsible for building the human perception area detection module, which includes manual parsing, extracting masks from people/backgrounds for editing, and optional mask refinement.
Xinyu ZHANG is responsible for local editing functions in the workflow of person-aware fashion editing, such as colour palette recalibration, pattern printing transfer, background replacement, and mask composition.
Langqi ZHONG is responsible for building the interactive Gradio system and providing project integration assistance, such as interface interaction workflow design, model and environment construction, local and LAN running scripts, testing, documentation, etc.

Zhengda SI (2230024178) **signature** 司政达 **date** 2026.5.21

Kai WEN (2230034054) **signature** 温恺 **date** 2026.5.21

Yiyi LIN (2230034031) **signature** 林易颖 **date** 2026.5.21

Xinyu ZHANG (2230034061) **signature** 张欣瑜 **date** 2026.5.21

Langqi ZHONG (2230034066) **signature** 钟浪颀 **date** 2026.5.21

Date: 2026.5.21

AI Usage Declaration

We, the undersigned group members, confirm that we have read and understood the course requirements, BNBU's academic integrity policies, and instructor guidelines regarding the use of generative AI tools (including but not limited to ChatGPT, Gemini, Doubao and other AI-assisted coding, problem-solving, or content-generation tools) in the completion of this group project.

We hereby declare the following:

1. **Declaration of AI Usage (Please select one of the following options)**

☑ We used generative AI tools in this project (coding, debugging, writing, or brainstorming).

☐ We did NOT use any generative AI tools in this project. All code, design, and written content are the original work of the group members.

If you checked "did NOT use," skip to Section 4: Signatures.

If you used AI, first tick the usage and then complete all sections below.

☑ Code generation, debugging, or optimization

☑ Explanation of algorithms, data structures, or technical concepts

☐ Drafting of written sections (introduction, methodology, results, discussion, etc.)

☑ Proofreading, editing, or formatting of text/code

☑ Literature review or research summary assistance

☐ Other (please specify): _____

2. **Detailed Disclosure of AI Usage:**

AI tool	**Primary User**	**Specific tasks**	Where it appears/ Impact	**Human contribution** (how you use it)
ChatGPT4.0	Xinyu ZHANG	Generating an image of portrait	Generated content image as example of style transfer	Used the generated image to test the result of our system

Chatgpt5.5	Xinyu ZHANG	Checking grammar	Assisted in revising the case analysis part of the paper	Made sure that the revised grammar was correct and that the original sentence was not changed
Chatgpt5.5	Xinyu ZHANG	Suggesting of Paragraph Titles	Gave several suggestions about how to name the scenarios	Selected and revised the name of each name
Chatgpt5.5	Xinyu ZHANG	Generated boilerplate code for mask composition pipeline	Used in initial implementation to speed up integration of multiple image masks	Reviewed and adapted the generated code to fit the local editing workflow
Chatgpt5.5	Xinyu ZHANG	Debugged segmentation alignment error in mask composition	Helped identify root cause in code logic	Applied the fix and verified the results on multiple test cases
Chatgpt5.5	Xinyu ZHANG	Helped write detailed docstrings and comments for complex editing modules	Improved code readability for the team	Adjusted documentation to reflect accurate algorithm details
Gemini	Wen KAI	Idea generation	Give me some ideas for the paper	For reference, it was not copied verbatim
Gemini	Wen KAI	Understanding the formula	Explain the meaning of each parameter in the formula	Understand and think
Gemini	Wen KAI	Check the details	Determine whether the content in the paper is consistent with that in the code	Identify shortcomings and adopt suggestions
Gemini	Wen KAI	Search for relevant literature	Search for literature related to VGG and Stable Diffusion	Partially adopted
Gemini	Wen KAI	Understanding model	Know the model architecture of Stable Diffusion and some of its mathematical principles	Have a rough understanding, but still need to combine it with the thesis
Gemini	Wen KAI	Understanding the paper	Help understand the structure diagram of Stable Diffusion	Understand and adopt
Gemini	Wen KAI	Search for papers	No papers based on the fashn-human-parser model were found	Search for open-source websites to obtain information
Gemini	Wen KAI	Ask about professional terms	Information about the binary mask is provided	Read carefully and understand
Gemini	Wen KAI	Ask about professional terms	Information about the dilation mechanism is provided	Read carefully and understand

Gemini	Wen KAI	Ask about professional terms	Information about the Pixel overflow is provided	Read carefully and understand
Gemini	Wen KAI	Generate an image	Flowchart for generating the VGG network	Not used, only for understanding
Gemini	Wen KAI	Explanatory function	The reason for providing the frozen network parameters of VGG	Read carefully and understand
Gemini	Wen KAI	Understanding framework	Explain the functions of Noise Injection and Denoising	Understand and adopt
WPS AI	Wen KAI	Dealing with garbled characters	The garbled characters in the formula have been dealt with	After checking for errors and using
ChatGPT5.5	Zhengda SI	Code explanation	Understand VGG-19 feature extraction	For reference, complete the coding
ChatGPT5.5	Zhengda SI	Clarification of idea	Clarify the loss of style and content	Apply these ideas to our own model design
ChatGPT5.5	Zhengda SI	Debugging support	Help check the NST optimization logic	Verify and modify the code
ChatGPT5.5	Zhengda SI	Parameter discussion	Propose the influence of iteration and loss weights on the results	Refer to the discussion, test and select parameters
ChatGPT5.5	Zhengda SI	Image generation	Generate the display diagram of 3D digital design	Understand and apply it in my presentation
Claude 4.5	Langqi ZHONG	Drew GUI wireframes and interface flow diagrams for early UI concept visualization	Used in team discussions and supervisor presentations to communicate the three-tab layout (NST, Local Edit, Img2Img)	Redesigned the actual Gradio interface based on team feedback
Claude 4.5	Langqi ZHONG	Generated PowerShell boilerplate for run_local.ps1 and run_lan.ps1 deployment scripts	Appears in scripts/ directory; enables one-click local and LAN server launch	Reviewed and modified scripts to ensure path compatibility with nn/config.py and correct offline mode behaviour
Claude 4.5	Langqi ZHONG	Produced initial Gradio tab-layout code for the three functional modules	Used in nn/gradioFinal.py as the structural foundation for the web UI	Check all event callbacks and backend bindings to match the actual VGG, diffusion, and parser pipelines

ChatGPT5.5	Yiyi LIN	Idea refinement / direction selection	Helped explore possible improvement directions for the coarse ROI-based method	Compared and studied each direction and finally selected the person-aware local appearance editing branch as the main improvement direction
ChatGPT5.5	Yiyi LIN	Technical research / reference search	Suggested possible methods related to human parsing, person-aware region detection, and mask-based local editing	Reviewed and evaluated the suggested resources, then selected the method suitable for the system goal
ChatGPT5.5	Yiyi LIN	Code understanding, debugging support	Assisted in checking code logic, identifying potential bugs, and suggesting modifications	Implemented and adjusted the code, tested the output results, and verified the final implementation
ChatGPT5.5	Yiyi LIN	Report proofreading and logic checking	Helped review whether the report explanations and technical choices were logically consistent and accurate	Checked the suggestions against the actual code and project process, corrected inaccurate claims, and finalized the report content manually

Table 1: AI Tool Declarations

3. Affirmation of understanding and responsibilities

By signing below, we affirm the following statements regarding our use of AI in this project:

(a) ☐ The core logic, architecture, and algorithmic design of the project are our own.

(b) ☐ We understand and can explain the content and every line of code submitted, regardless of whether it was AI-assisted.

(c) ☐ All AI-generated content has been fully reviewed, verified for accuracy, and adapted by us. No AI tool was used to generate **entire** modules or core functionalities without our significant modification and understanding.

(d) ☐ We have clearly indicated within the project materials (e.g., **code comments**, README file, in-text citations, or appendices) the specific sections/elements where AI tools were used, in accordance with instructor requirements.

(e) ☐ We take complete responsibility for the correctness, originality, and academic integrity of the final submitted work, including any errors or inconsistencies in AI-generated content that we failed to identify or correct.

(f) ☐ This submission adheres to the BNBU's academic integrity policy and the course's specific rules on AI tool usage. We understand that intentional misrepresentation or failure to truthfully declare AI usage may result in penalties for academic misconduct.

4. Group member information

Full Name	Student ID	Signature (Handwritten or digital)	Date
Zhengda SI	2230024178	司政达	2026.5.21
Kai WEN	2230034054	温恺	2026.5.21
Yiyi LIN	2230034031	林艺乘	2026.5.21
Xinyu ZHANG	2230034061	张欣瑜	2026.5.21
Langqi ZHONG	2230034066	钟朗琦	2026.5.21

Table 2: Group Member

ABSTRACT

The transfer of traditional styles can achieve a transformation at the artistic level of images, but it still has limitations in terms of visual details and personalized design. To address these issues, this project aims to develop an interactive and controllable system for image style transfer and local editing. This system will build a new framework that integrates traditional style transfer models, the stable diffusion model, and a person-aware local editing module. Firstly, the neural style transfer technology based on VGG-19 is used to complete the traditional style transfer, and then the diffusion model can be optionally used to further enhance its visual expressiveness. In addition, the segmentation mask of the person-aware local editing module is applied to achieve customized editing of specific clothing areas, specifically including colour changes, pattern style transitions, and clothing colouring. Gradio is adopted to achieve interaction with users, including uploading images, selecting modes, and choosing areas. Overall, this project has further improved the integration of traditional style transfer and diffusion models, and on this basis, has achieved personalized local editing functions, building a more reliable, flexible, and controllable artificial intelligence image style transformation system.

1 Introduction

1.1 Background

Digital media and creative industries as a whole are being impacted by advanced image editing technologies. Neural Style Transfer (NST) combines and "styles" image content using a trained neural network. Contemporary models based on diffusion outperform NST when it comes to style generation. However, when both techniques are applied globally, they often affect several other aspects of an image. In this paper, we design a system for image style transfer and local editing using Gradio software. Our system employs an integrated pipeline combining torchvision VGG19, Stable Diffusion v1.5 Img2Img, and the fashn-ai human parser.

1.2 Project Objectives and Research Questions

This project aims to create a system to edit the style of images locally and interactively. Our primary research question concerns the integration of semantic segmentation masks with one of two approaches: the first being diffusion models, and the second being optimization-based approaches. More specifically, we ask the questions of how clothing or background as a style can be changed using segmentation masks, and how style changes can be confined to specific regions that the user selects. A key requirement is that during the editing process, all unselected pixels are entirely preserved.

1.3 Significance and Application Scenarios

The system intelligently manages the coordination of several collaborative models in computer vision. We expect it to serve directly in digital media production, prototyping in fashion design, e-commerce 'look-book' production, and others. Users can execute three high precision local editing operations: palette recolouring, shifting a print pattern and style-based background diffusion. Each edit is contained within the safe semantic mask, ensuring the rest of the area is left intact and improving the overall controllability and efficiency of the image synthesis process.

2 Related Work

2.1 Neural Style Transfer

Neural Style Transfer (NST) modifies an image to mimic the appearance of another by using convolutional networks to analyse feature statistics. The optimization-based framework relies on VGG19. Gatys et al. [1] extract content features at higher levels and style features using Gram matrices, and then generate the target image using gradient descent, which creates stylized images at higher quality, but takes longer to implement. NST was later optimized using feed-forward networks [2] and adaptive normalization [3], which prioritized speed over quality. We integrate the original VGG19 NST [1] within a diffusion-based framework to preserve the authenticity of the stylization.

2.2 Project Objectives and Research Questions

Currently leading the way for controllable image generation are diffusion models. One significant example is Stable Diffusion v1.5 [4]. It compresses images into latent space utilizing a variational autoencoder, which speeds up the process of high-resolution image generation. It also features an Img2Img pipeline, which applies denoising diffusion to a noised version of an image while processing a text prompt. Here, the strength parameter is a trade-off between how faithful the output image is to the input, and how creative the output is. When editing images, inpainting can often negatively affect the structures of images that encompass fine details such as hair or fabric, especially when inpainting directly. Because of that, we prefer the Img2Img pipeline. It will create a highly stylized version of the image. Then we combine only the background of the new image with the original photograph, using a safe semantic mask. This will make sure no edits are made to fine structures.

2.3 Significance and Application Scenarios

Human parsing divides a portrait into segments like hair, clothing, accessories, and background. In the past, methods depended on features delineated by hand. Nowadays, methods depend on deep convolutional networks that are trained on fashion datasets that are labelled with dense annotations [5], [6]. Our system uses the fashn-ai/fashn-human-parser, a model that is tailored toward fashion and portrait scenarios. The model provides a labelled image and segmented masks for a single piece of clothing which are the spatial limitations for all local editing operations—colour palette editing, pattern transfer, and background swapping. Precise parsing of semantics replaces tedious manual mask annotation, which is critical for users without expertise in this area.

2.4 Significance and Application Scenarios

Stable Diffusion WebUI and ComfyUI are examples of open-source systems for generative editing. Though they provide detail and control for generative editing, they are made for general editing. These systems are versatile, but that comes with the cost of steep and complex systems that are not

user-friendly for those not versed in prompt engineering. Our system, based on Gradio, has a different approach. It is task oriented and has three separate tabs for VGG Style Transfer (NST), Person-aware Local Edit, and Stable Diffusion Img2Img. Each tab has an entire editing process behind it but provides a simple interface to maximize ease of access for users.

3 Methodology

3.1 Overall System Design

The system is designed as an interactive framework for image style transfer and local appearance editing. Rather than relying on a single model, it adopts a multi-branch design to explore different levels of stylization and controllability. Its overall goal is to support visual style exploration by combining explicit neural style transfer, diffusion-based image editing and refinement, and mask-based local editing.

To achieve this goal, the system is organized into three main branches.

The first branch focuses on global image stylization. It takes a content image and a reference style image as the main inputs. A VGG-based Neural Style Transfer (NST) module first transfers visual style from the reference image to the content image. The intermediate result can then be refined with Stable Diffusion image-to-image (Img2Img) under text-prompt guidance. In this two-stage workflow, the style image mainly affects the result through the first stage, while the diffusion stage is mainly used to improve visual quality, enrich fine details, and provide additional prompt-based control.

The second branch is designed for cases in which only a content image is available. In this setting, the system uses Stable Diffusion Img2Img to edit the input image directly from text prompts and user-defined parameters. This branch therefore supports prompt-guided visual editing without requiring a separate reference style image. To make the prompting process easier and more consistent, the interface also provides preset prompt options for common visual styles.

The third branch focuses on local appearance editing for person images. This branch is motivated by a limitation of global stylization: users may want to change only clothing regions or the background while preserving the face, skin, and overall body structure. To address this need, the system introduces a person-aware editing workflow based on semantic masks. It detects person-related semantic regions, allows users to select target areas, and supports local editing modes such as recolouring, pattern-like transfer, and background editing.

Taken together, these branches form a style-oriented image editing system that supports both global stylization and additional spatial control.

3.2 System Architecture

The system follows a multi-branch architecture built around an interactive Gradio interface. Figures 1–3 illustrate the three processing branches of the system and their corresponding data flows. Each branch starts from user-provided inputs and produces a final edited image through a different sequence of modules.

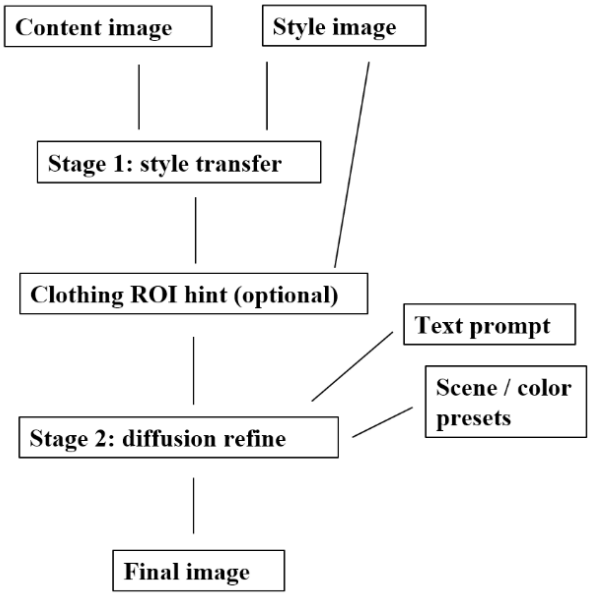


Figure 1: Branch 1: Global style transfer and diffusion refinement

The first branch, shown in Figure 1, performs global style transfer followed by diffusion refinement. A content image and a reference style image are first processed by the Neural Style Transfer module to produce a stylized intermediate result. In clothing-oriented scene settings, an optional ROI hint can be introduced before the refinement stage to provide weak spatial guidance, encouraging the diffusion model to focus more on clothing-related regions rather than other parts of the image. The intermediate output is then passed to the diffusion refinement stage together with a text prompt and optional scene or colour presets. In this branch, the style image is mainly used in the first stage, while the diffusion stage refines the intermediate result under prompt and preset guidance.

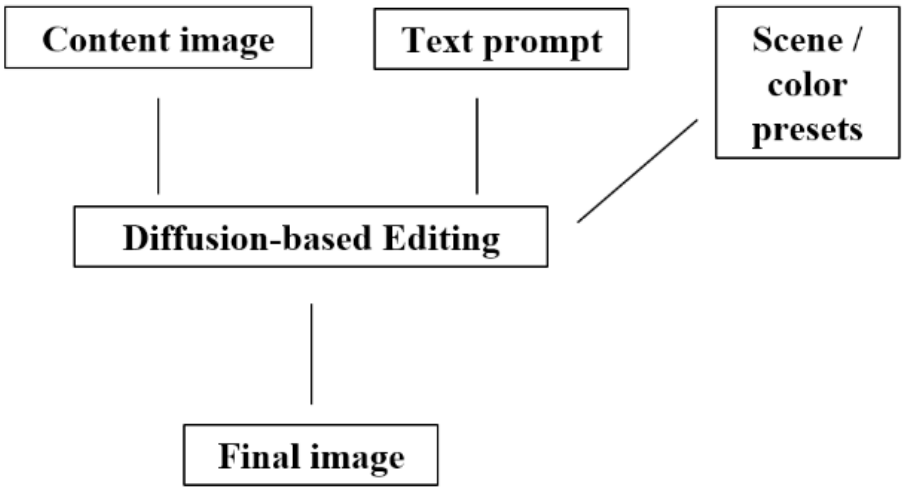


Figure 2: Branch 2: Prompt-guided diffusion editing

The second branch, shown in Figure 2, represents the direct diffusion-based editing mode of the system. It takes a content image, a text prompt, and optional scene or colour presets as inputs, and

produces the final image through Stable Diffusion Img2Img. This pathway is more suitable for users who prefer a straightforward workflow based on a single input image and text guidance.

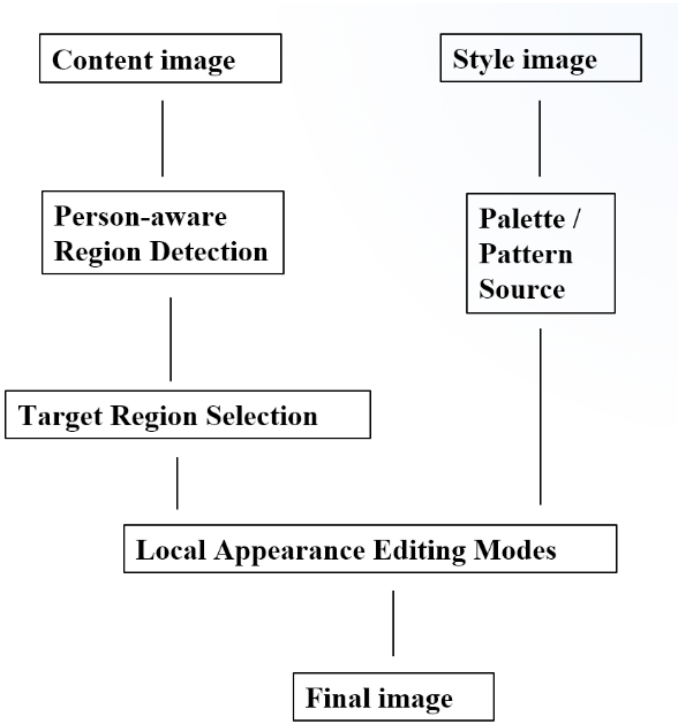


Figure 3: Branch 3: Person-aware local appearance editing

The third branch, shown in Figure 3, implements person-aware local appearance editing. The content image is first processed for region detection, after which the user selects a target area, such as clothing or background regions. Depending on the selected editing mode, the style image provides palette or pattern cues for local editing. The final result is then produced through recolouring, pattern-like transfer, or background editing. Unlike the first two branches, this pathway includes explicit region detection and target selection before editing.

After defining the three-branch architecture, next section describes the core computational modules used in these branches. These modules include the VGG-based NST module, the Stable Diffusion Img2Img module, and the person-aware semantic segmentation module.

3.3 Model

3.3.1 Visual Geometry Group (VGG)-based Neural Style Transfer Module

Network Architecture and Feature Layers

This system uses the VGG-19 convolutional neural network pre-trained on ImageNet as the backbone architecture for visual feature extraction:

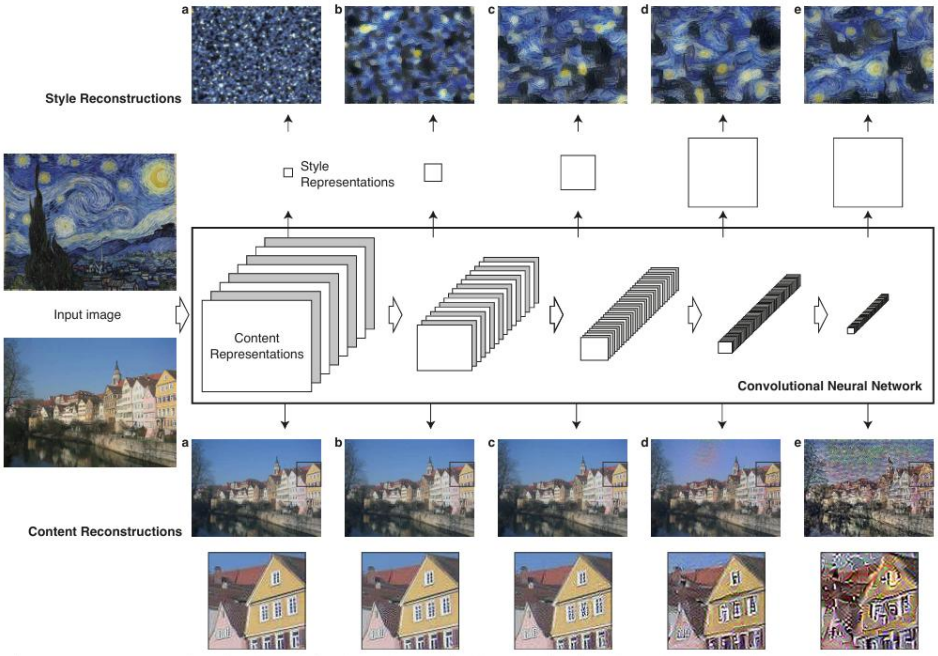


Figure 4: Image representations in a Convolutional Neural Network (CNN) [1]

In the actual algorithm deployment of this module, all network weights of VGG-19 are completely frozen (i.e., gradient backpropagation is blocked) and only used as static nonlinear feature mapping operators to ensure that the extracted high-level visual feature space remains stable.

In order to effectively separate the semantic content and artistic style of the image, the feature map output by the network is extracted hierarchically. The system specifies the deep convolutional layer (such as the fourth convolutional block (conv_4)) as the content feature layer and uses its high-level receptive field to anchor the macroscopic geometric structure of the original image. At the same time, the system selects multiple continuous convolutional layers from shallow layers to deep layers to form a set of style feature layers, so as to extract the comprehensive texture information from micro strokes to macro tones in the reference image at multiple scales.

Core Loss Functions

The stylistic driving force of modules stems from the joint optimization of content loss and style loss. The system defines the total optimization objective as the weighted sum of the normalized content loss and style loss:

* Content Loss ($\mathcal{L}_{content}$): It is obtained by calculating the mean square error (MSE) between the generated image and the feature tensor output by the fourth convolutional layer (conv_4) of the content reference image. To prevent the content weight from dominating the gradient, the system normalizes this error by dividing it by the number of content layers and multiplying it by a preset content weight coefficient to constrain the structural fidelity of the output image.

* Style Loss ($\mathcal{L}_{style}$): To extract the texture features with spatial invariance, the system

calculates the Gram Matrix of the style feature map. Gram matrix quantifies the texture and color correlation of an image by calculating the inner product between different feature channels. The style loss is the mean squared error between the Gram matrix of the generated image and the style reference image.

* Total loss ($\mathcal{L}_{total}$): The final objective function is a linear combination of the content weighting coefficient α and the style weighting coefficient β (α and β are user-adjustable hyperparameters):

$$\mathcal{L}_{total} = \alpha\mathcal{L}_{content} + \beta\mathcal{L}_{total}$$

Pixel-Level Optimization

This Project is different from the parameter training paradigm of traditional deep learning models, the optimization goal of this module is not to update the weights of the neural network itself. Instead, the system explicitly declares the pixel tensor of the input image as the core variable to be optimized.

In the whole iterative optimization cycle, the system uses the error back propagation mechanism to calculate the gradient of the total target loss with respect to the image pixels and uses the Adaptive Moment Estimation (Adam) optimization algorithm to directly iteratively update the pixel values of the image. To ensure that the generated image tensor remains in the legal physical colour space (normalized [0, 1] interval) after multiple gradient superpositions, the system introduces a numerical Clipping mechanism at the end of each single-step iteration to force correction of any overflowing abnormal pixels.

This module of the multi-level image editing system is used to set up the basic semantics and tone. Quickly build an "artistic skeleton" with good texture and a clear structure in pixel space. This output is then used as a high-quality initial value for the following Stable Diffusion model pipeline to enhance the quality of the resulting image and prevent structural collapse of the diffusion model that may occur due to large redraws.

3.3.2 Latent Diffusion Image-to-Image Module

Network Architecture and Latent Space Decomposition

To achieve high-fidelity global style transfer and image quality refinement in artistic image editing, this system integrates a Stable Diffusion generation engine based on the Latent Diffusion Models (LDM) architecture [4]. The traditional diffusion model directly performs the calculation in the high-dimensional pixel space, which consumes huge resources. To break through this bottleneck, this module introduces the latent space decoupling mechanism, and its core architecture is composed of three sub-networks:

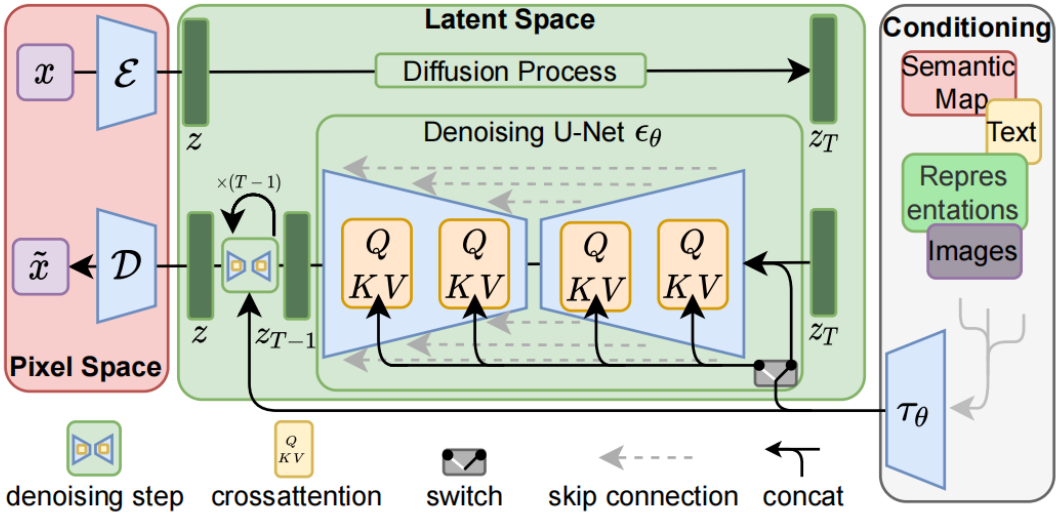


Figure 5: The structure of Latent Diffusion Models [4]

* Variational Autoencoder (VAE): Consists of an encoder $\mathcal{E}$ and a decoder $\mathcal{D}$. The encoder is responsible for compressing the high-dimensional pixel image x into a low-dimensional continuous latent representation space $z = \mathcal{E}(x)$, so as to strip the high-frequency pixel-level redundant information. At the end of the generation pipeline, the decoder inverts the optimized latent space variables back into pixel space.

* Conditional Denoising U-Net: As the backbone of the generation engine, this network operates in a low-dimensional latent space and is responsible for predicting and stripping noise residuals from the hidden variables at each time step t.

* Text Encoder: The system employs a pre-trained Contrastive Language-Image Pre-training (CLIP) model [7] to transform user-input natural language prompts into continuous semantic conditional vectors $\tau(\mathcal{P})$. These vectors serve as cross-modal conditional constraints to guide the denoising trajectory of the U-Net.

Mathematical Principles of Controlled Diffusion

The Image-to-Image (Img2Img) algorithm paradigm adopted in this module is essentially a combination of a controlled Markovian forward perturbation process and a conditional reverse generation process.

* In the forward noise injection stage, the system does not start from pure Gaussian noise but takes the latent representation z_0 of the original input image as the initial state. According to a specific noise scheduling strategy, the algorithm injects an appropriate amount of Gaussian noise $\epsilon \sim \mathcal{N}(0, \mathrm{I})$ into z_0. At any time step t, the analytical expression of the noisy latent variable z_t is [8]:

$$z_t = \sqrt{\bar{\alpha}_t} z_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon$$

$\bar{\alpha}_t$ is the cumulative product of the noise variance.

* In actual Image-to-Image applications, the cut-off time step t_0 for forward noise addition is strictly controlled by the system's redraw amplitude parameter (Strength, $\eta \in [0,1]$) [9]. Let the total denoising sampling steps be T, then the initial noise injection step size is defined as:

$$t_0 = \lfloor \eta \times T \rfloor$$

This formula indicates that when η is small, the system only injects a tiny amount of disturbance, and the network will highly preserve the physical structure of the original image during the reverse denoising process; when η approaches 1, the latent variable will be completely destroyed and degenerate into a pattern similar to a purely random text-to-image generation.

* In the reverse denoising stage, the U-Net integrates the text-conditioned features into the image generation process through a cross-attention mechanism (Cross-Attention) within its architecture. The formula for calculating the attention is [10]:

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d}}\right)V$$

Among them, the query vector Q is derived from the deep feature map of the image latent space, while the key vector K and value vector V are mapped from the feature vector$\tau(\mathcal{P})$ of the text prompt. Through this mechanism, the system can precisely render the text description onto the corresponding image region in the spatial dimension.

Main Functions in the System

* Break through the limits of traditional image quality: Directly use the module for images that have been damaged by pixel noise and distortion introduced in the front-end VGG NST as the initial latent variable z_0. U-Net is used to remove the defects of traditional methods, reconstruct the material appearance of real objects, add realistic volumetric shadows, and so on; thus, it is much more realistic visually.

* Balance Structural Fidelity and Artistic Generation: Quantify and reveal the re-painting amplitude parameter η (Strength) to constrain it within a lower range during editing. The final generated image will be in a new style, but the face and body of the characters will be kept intact; the structure will not be altered.

* Context-aware prompt word intervention: To prevent the open-ended nature of generation from drifting, positive scene constraints (e.g., "detailed garment texture") and negative penalty terms (e.g., "no change to the face") are automatically added to the cross-attention layer (Cross-Attention) dynamically. The above Design will enable the underlying diffusion model to meet the requirements of the "person-image local appearance editing" in this project and achieve controllable generation results.

3.3.3 Person-aware Semantic Segmentation Module

Network Architecture and Dense Prediction Mechanism

To achieve precise "local perception editing" of clothing and background in images, the system must have the ability to understand the spatial topology of the human body at the pixel level. For this purpose, this module integrates the fashn-human-parser semantic segmentation model, which has been fine-tuned specifically for fashion and human parsing tasks. Unlike traditional object detection models that only output target bounding boxes, this parser is based on the advanced SegFormer architecture [11], featuring a dense prediction architecture composed of a hierarchical Transformer encoder and a lightweight fully connected (MLP) decoder.

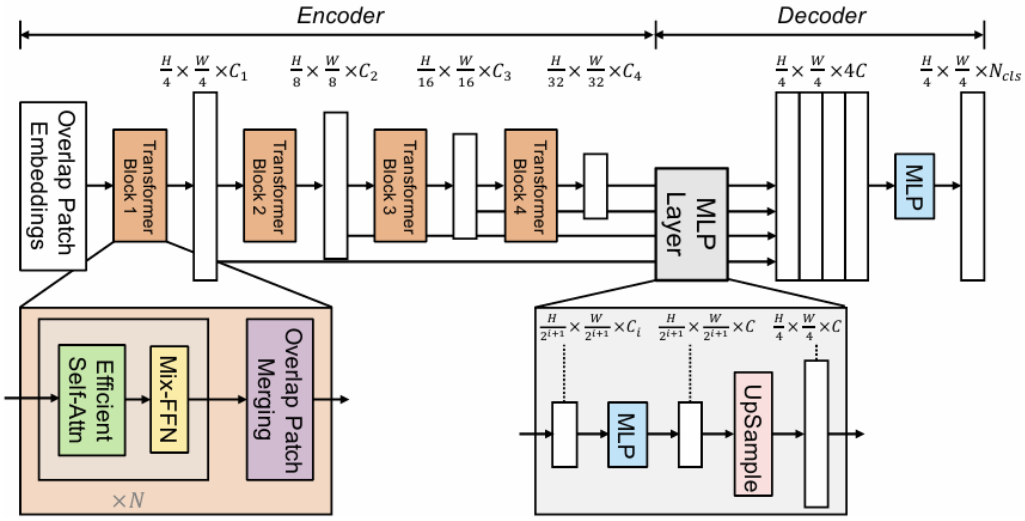


Figure 6: The proposed SegFormer framework [11]

During the reasoning stage, the model maps the high-dimensional input image to a deep feature pyramid and performs a pixel-wise classification task (Pixel-wise Classification). This model can precisely map each pixel in the image space to a predefined discrete semantic set. In this project, the semantic space is strictly divided into 18 fine categories, covering background, face, hair, as well as upper body clothing, pants, and other detailed human anatomical parts and clothing accessories, thereby providing the system with extremely high-resolution semantic understanding.

Mask Aggregation and Morphological Safety Algorithm

To convert the discrete semantic classification map output by the model into spatial constraints that can be directly used by the subsequent generation model, the system has built a dedicated mask extraction and morphological processing pipeline at the bottom layer. First, the system will traverse and filter the original semantic classification map based on the editing target specified by the user (for example, only editing the upper garment and pants). The system extracts all pixels belonging to the target category and sets non-target pixels to zero, thereby aggregating to generate independent

binary masks (Binary Masks).

Secondly, when performing background redrawing, to prevent the diffusion model from damaging the edges of the characters, this module introduces the "Dilation" operator from morphological operations. The algorithm uniformly expands the basic mask of the characters outward by several pixels, artificially creating a physical buffer zone. Then, the inverse of the expanded contour is taken to obtain the safe background mask. This mechanism can effectively separate the area that needs to be redrawn from the character entity, fundamentally avoiding the situation of pixel overflow when generating the background.

Core Engineering Role in the Pipeline

In a multi-stage image editing pipeline, this semantic segmentation module plays a crucial role in spatial positioning and boundary constraint. First, precisely extract high-resolution pixel-level masks to separate the subject from the target editing area (such as clothing or background), and clearly divide the original image into feature-retaining areas and local redrawing areas. This step sets a tight spatial bound for the following Stable Diffusion model and thus avoids boundary overflow and image drift in the diffusion process. A morphological dilation module has been added to the generative model at the junction of mask regions to deal with the problem of seam artifacts. At the end of the pipeline, in conjunction with the Edge smoothing operation, a smooth visual transition is achieved between the newly generated elements and the actual skin tone of the subject to ensure the structural integrity and output quality of the final image at the algorithmic level.

3.4 Mode Routing and Inference Workflow

Building on the modules described above, this section explains how the system routes user inputs to different inference workflows.

The runtime workflow of the system is mainly determined by the selected tab in the Gradio interface. The interface contains three top-level workflows: VGG Style Transfer (NST), Person-aware Local Edit, and Stable Diffusion Img2Img. Each tab corresponds to a different execution path and requires a different set of inputs.

In the VGG Style Transfer (NST) tab, the user provides a content image and a reference style image, together with NST parameters such as the number of iterations and the content and style weights. The system first runs the VGG-based Neural Style Transfer stage and produces a stylized intermediate result. If diffusion refinement is disabled, this result is returned directly. If diffusion refinement is enabled, the intermediate result is passed to Stable Diffusion Img2Img together with the selected prompt settings and diffusion parameters, such as strength, sampling steps, and guidance scale. As a result, this tab supports both a single-stage NST workflow and a two-stage NST-plus-diffusion workflow. In clothing-oriented use cases, an optional ROI hint may be introduced before the

refinement stage as weak spatial guidance.

In the Stable Diffusion Img2Img tab, the workflow is more direct. The user provides a content image, prompt settings, and diffusion parameters, and the system sends the content image directly to Stable Diffusion Img2Img. This workflow directly edits the input image under prompt guidance and serves as the most straightforward diffusion-based editing path in the system.

The Person-aware Local Edit tab follows a step-based workflow. After the user uploads the content image, the system first performs person-aware region detection after the user clicks Detect Person Regions. The detected semantic regions are then displayed for target selection. The next routing step depends on the selected editing target. If the target is a person-related region, the system enables region selection for local editing and routes the request to palette recolouring or pattern-like transfer. In this case, the style image provides palette or pattern cues for the selected region. If the target is background, the system switches to background-related controls and routes the request to solid-colour replacement, direct style-image replacement, or style-diffused background generation. In this route, the style image may serve directly as the replacement background or as a visual reference for diffusion-based background generation, while the solid-colour mode can operate without a style image.

Mask-based control is central to the execution of the person-aware branch. For person-related editing, the selected semantic regions are merged into an effective editing mask, and the local modification is restricted to that area. For background editing, the system uses a safe background mask that excludes pixels near the person boundary. This protects the subject while allowing the background to be replaced or stylized. The final output is therefore produced through branch-specific inference followed by mask-guided composition when needed.

Overall, the system uses different inference paths for different editing tasks and applies mask-based composition when local control is required.

3.5 Technical Choices and System Rationale

Building on the routing workflow described above, this section explains why the system adopts this combination of models, interface design, and region-control mechanisms. The technical design follows a task-decomposition strategy rather than a single-model strategy. Instead of training new models from scratch, the system integrates pretrained components, and each module is selected to address a different requirement of the editing task. In this design, explicit style transfer is separated from prompt-guided refinement, while spatial control is handled by region-aware local editing. This decomposition makes the system easier to organize, explain, and deploy within the scope of the project.

A VGG-based Neural Style Transfer (NST) module is used because it provides an explicit and

interpretable way to incorporate reference-style information from a style image into a content image [1]. In this branch, the role of the style image is clear: it directly contributes to the stylization stage rather than serving only as an indirect visual reference. This is useful in the current system because the project began with classical feature-based style transfer and then extended toward diffusion refinement. In addition, the VGG-based NST workflow does not require retraining and is suitable for a modular two-stage design.

Stable Diffusion Img2Img is used because it provides a pretrained generative model for prompt-guided editing and refinement without requiring model training [4]. In the hybrid NST-plus-diffusion workflow, Stable Diffusion mainly refines the NST intermediate result and improves visual quality, fine visual details, and prompt-based controllability. In the standalone diffusion branch, it serves as a direct image editing module driven mainly by the content image and text prompt. This makes it a practical choice for the project because it offers a balance between generation quality, parameter control, open-source accessibility, and local deployment feasibility.

The person-aware branch is introduced because global stylization alone does not provide sufficient spatial control for person images. In fashion-oriented or portrait-related editing, users may want to modify only selected regions, such as clothing, accessories, hair, or the background, while preserving the face and body structure. The system initially explored ROI-style guidance as a coarse spatial cue, but this approach could not accurately follow real human and clothing boundaries. Therefore, the system extends this idea into semantic-mask-based local appearance editing. Compared with rough ROI-style guidance, semantic masks derived from fashion-oriented human parsing align more naturally with person-related regions and provide more precise editing constraints [12]. This makes mask-based local editing more suitable for controlling where appearance changes should be applied.

Practical deployment considerations also influenced the technical choices. The system uses Gradio as an interactive front end because it supports rapid integration of image inputs, parameter controls, and step-based workflows in a single Python environment [13]. In addition, the project includes model download scripts, cache-path management, and local or LAN launch settings to improve reproducibility and reduce deployment ambiguity across machines. These choices support the goal of building not only a research prototype, but also a usable and reproducible interactive editing system.

4 Case Analysis and Discussion

In this section, we design and discuss three application cases for our system: global style transfer, fashion editing for people, and artistic typography generation. In each case, we first explain the main task, and then use our example results to show how the system works and then discuss the limitation within this area.

4.1 Case 1: Global Style Transfer

Global style transfer changes the overall visual style of an image while keeping the main content easy to recognize. It works well for transforming portraits and everyday photos into cinematic, anime, watercolour, oil painting, or vintage styles, for example landscapes, portraits, campus photos, and poster materials. While the main problem in this scenario is how to keep enough original content while making the new style clear.

Figure 7 shows four examples of global style transfer. The first example uses a Zhangjiajie landscape photo as the content image and a Chinese shanshui painting as the style image. In this case, when the content image is uploaded, VGG 19 keeps the mountain shapes, sky area, and general composition through content loss. At the same time, Gram Matrix style loss adds the ink like colour and brush texture from the shanshui painting. The second example uses an AI generated blonde female portrait as the content image, and the target style is Picasso style. We can set the prompt "Picasso style abstract portrait" manually in our webpage to achieve that. This example is closer to prompt guided stylization. Img2Img strength in this scenario controls how much the model changes the original portrait. A lower strength keeps more of the face layout and hair shape, while a higher strength creates stronger geometric shapes, colours, and abstraction.

The third and fourth examples show that the same method can be used in other scenes. The third example uses a BNBU library photo as the content image and Van Gogh's Starry Night as the style image. As a result, the BNBU library photo becomes an impressionist style image after generation. The fourth example uses an Amsterdam canal photo as the content image and a digital fractal mandala pattern as the style image, creating a more decorative result. We can see that the structure of the original image is well preserved, and the appropriate texture is added to improve the texture of the whole picture. These examples show that global style transfer can be used for landscape design, campus publicity, portrait stylization, and experimental visual design.

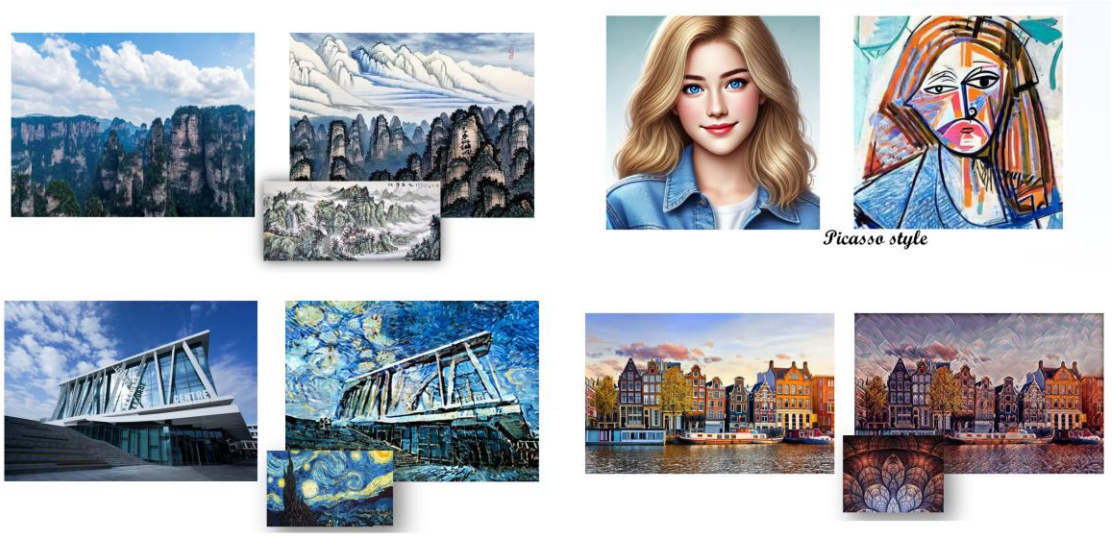


Figure 7: Global style transfer examples with four content and style settings based on [14], [15], [16], [17], and [18]

4.2 Case 2: Person-Aware Fashion Editing

Fashion editing for people is different from global style transfer because only selected parts should change. In this case, the system edits clothing or background while keeping the face, pose, body shape, and other objects stable. This is closer to real fashion editing, because changing the whole image would not be useful. The implementation follows the local edit module in the module documentation. In this case, the content image and reference image will be sent to the human parser, which uses fashn ai fashn human parser. The parser produces a label map, garment masks, a selected mask, and a safe background mask. These masks tell the system which pixels can be edited, and which parts should stay unchanged. Details of this method are fully mentioned in the methodology part.

For example, we use one fashion model image and two polka dot pattern images. The blue polka dot pattern is applied to the upper garment, while the multicolour polka dot pattern with different dot sizes is applied to the skirt. As we can see, through semantic segmentation, every part of the image can be segmented with great accuracy, even the waist belt of the figure can be separated from the clothes with great accuracy. It is notable that the dots are not just a simple overlay or map but can be adjusted according to different parts of the clothing, light and shadow. The application of this scene can not only provide more inspiration and rapid implementation methods for fashion design, but also greatly reduce the workload of designers, so that designers can quickly see the demo of design ideas and spend more energy on pure inspiration.

The main limitation is mask accuracy. If the parser includes hair, hands, belts, or bags inside the garment mask, the pattern may appear in the wrong area. If the mask misses' part of the clothing, the print will look incomplete. Optional SAM2 refinement can improve the boundary when the checkpoint is available, but manual mask correction would still make this workflow more reliable.



Figure 8: Fashion editing examples for the person image. Images are from [19], [20], and [21]

4.3 Case 3: Artistic Typography and Advertising Design

The third task we design is Artistic typography and advertising design. The goal of this scenario is not to transfer style between normal photos, but to create a useful visual design element. This case can be used for logo letters, campaign initials, poster graphics, or social media visuals. We hope that the result should look decorative, but the letter still needs to be readable. For demonstration, we use artistic letter B as the structural input and a decorative vector pattern as the style reference. The pattern is based on European acanthus scroll ornament, with purple butterfly elements as extra decoration. The generated result keeps the clear shape of the letter B, while adding the ornament inside the letter and around the border. Diffusion model then redesigns the whole layout, so the final image looks more like an advertising composition instead of just a textured letter.



Figure 9: Artistic typography example for the letter B, using acanthus scroll ornament and purple butterfly elements as decorative references. Images are from [22] and [23]

4.4 Comparative Discussion

The three cases are different mainly in editing scope and control. Global style transfer changes the whole image, so it is suitable for landscapes, portraits, and campus scenes. Fashion editing changes only selected masks, so it is better for garment and background edits. Typography generation is useful in the stage of early ideas for artistic designing. Imagine that when several design directions need to be compared quickly and drawing manually for each costs a lot of time. This system could help you to reduce manual work in this kind of early design stage. Now, we will further compare the differences between the three cases in terms of inputs, models, etc.

Different image generation and editing tasks have different inputs, methods, values, and limitations. In style transfer tasks, the system usually takes content images, style images, or text prompts as the main inputs. Methods such as VGG 19 neural style transfer and Img2Img strength control are commonly used. These methods can generate landscape images, campus scenes, portraits, and decorative drafts. However, editing the whole image at the same time may also change some important details in the original image. In fashion editing tasks, model photos and clothing pattern images are used as the inputs. Human parser masks and pattern transfer methods can help users preview upper garment and skirt designs. At the same time, mask detection errors may sometimes cause pattern leakage problems. For typography generation, the system mainly uses letter images and ornament references together with prompt guided diffusion generation. This application is useful for logo design and poster draft creation, although the generated text is still difficult to edit completely.

5 Conclusion and Future Work

In conclusion, this project has successfully designed an interactive and controllable style transfer and local editing system based on the style transfer model. A smooth operation of the system was achieved by using an image editing tool built with Gradio and a combination of different core workflows. Make the system capable of global style transfer, diffusion image refinement, and local style editing of characters' clothing.

The project integrates the traditional style transfer and the generative diffusion models, which allows the system to not only transfer the style of the relevant content features but also enrich the visual details of the results. Our project doesn't only carry out the style transfer for the whole picture, like the normal style transfer system. Rather, it can realize style transfer on a local part (character clothing) by the character area detection and the driving mechanism of the character clothing mask, and the system is more targeted and controlled. Moreover, and most importantly, this project has broken down the complex workflow into simple and easy-to-use tools by designing interactive pages, which further increases the practical value of the project.

At the technical level, the value of the system has been realized, and this realization has encouraged the system's commercialization and socialization, giving rise to the system's practical application value. This system can be used to help professionals, such as fashion designers, to reference this design using editing functions such as colour change and pattern transfer in specific clothing areas, which can be used for further exploration. This system allows users who aren't design experts to create individualized design styles in a more intuitive way, helping to spread the creation of digital art.

In the current system, local clothing refinement based on diffusion is mainly carried out by using masks and prompts, and style images are not strictly used as visual conditional signals. One direction that is promising is IP-Adapter, which will make style images no longer just for show, but will be able to send visual information from style images into diffusion to truly view style images. In addition, in the future, the system can no longer be confined to the two-dimensional level, and its image editing workflow should be further expanded to three dimensions. These innovations will further enhance the application value of the system in the field of design, making it more in line with the development of the digital design era.

References

[1] L. Gatys, A. Ecker, and M. Bethge, "Image Style Transfer Using Convolutional Neural Networks," 2016. Available: https://openaccess.thecvf.com/content_cvpr_2016/papers/Gatys_Image_Style_Transfer_CVPR_2016_paper.pdf

[2] J. Johnson, A. Alahi, and L. Fei-Fei, "Perceptual Losses for Real-Time Style Transfer and Super-Resolution." Available: https://arxiv.org/pdf/1603.08155

[3] X. Huang and S. Belongie, "Arbitrary Style Transfer in Real-time with Adaptive Instance Normalization." Available: https://openaccess.thecvf.com/content_ICCV_2017/papers/Huang_Arbitrary_Style_Transfer_ICCV_2017_paper.pdf

[4] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer, "High-Resolution Image Synthesis With Latent Diffusion Models," Thecvf.com, pp. 10684–10695, 2022, Available: https://openaccess.thecvf.com/content/CVPR2022/html/Rombach_High-Resolution_Image_Synthesis_With_Latent_Diffusion_Models_CVPR_2022_paper.html?utm_source=rns.dwaiai.de

[5] K. Gong, X. Liang, D. Zhang, X. Shen, and L. Lin, "Look into Person: Self-supervised Structure-sensitive Learning and A New Benchmark for Human Parsing." Available: https://openaccess.thecvf.com/content_cvpr_2017/papers/Gong_Look_Into_Person_CVPR_2017_paper.pdf

[6] P. Li, Y. Xu, Y. Wei, and Y. Yang, "Self-Correction for Human Parsing," IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. 44, no. 6, pp. 3260–3271, Jun. 2022, doi: https://doi.org/10.1109/tpami.2020.3048039.

[7] A. Radford et al., "Learning Transferable Visual Models From Natural Language Supervision," 2021. Available: https://proceedings.mlr.press/v139/radford21a/radford21a.pdf

[8] J. Ho, A. Jain, and P. Abbeel, "Denoising Diffusion Probabilistic Models." Available: https://proceedings.neurips.cc/paper/2020/file/4c5bcfec8584af0d967f1ab10179ca4b-Paper.pdf

[9] C. Meng et al., "SDEdit: Guided Image Synthesis and Editing with Stochastic Differential Equations," arXiv:2108.01073 [cs], Jan. 2022, Available: https://arxiv.org/abs/2108.01073

[10] A. Vaswani et al., "Attention Is All You Need," 2017. Available: https://proceedings.neurips.cc/paper/2017/file/3f5ee243547dee91fbd053c1c4a845aa-Paper.pdf

[11] E. Xie, W. Wang, Z. Yu, A. Anandkumar, J. Alvarez, and P. Luo, "SegFormer: Simple and Efficient Design for Semantic Segmentation with Transformers." Available: https://proceedings.neurips.cc/paper/2021/file/64f1f27bf1b4ec22924fd0acb550c235-Paper.pdf

[12] fashn-ai, "FASHN Human Parser," Hugging Face model card. [Online]. Available: https://huggingface.co/fashn-ai/fashn-human-parser. [Accessed: May 20, 2026].

[13] A. Abid, A. Abdalla, A. Abid, D. Khan, A. Alfozan, and J. Zou, "Gradio: Hassle-Free Sharing and Testing of ML Models in the Wild," arXiv preprint arXiv:1906.02569, 2019. [Online]. Available: https://arxiv.org/abs/1906.02569

[14] "Zhangjiajie", 588ku.com, 2026. https://588ku.com/so/45492617/ (accessed May 20, 2026).

[15] Shikai, "Taoyuanchunxiao", "Landscape Painting | Chinese Painting," China Online Museum. https://www.comuseum.com/painting/landscape-painting/

[16] "Library of BNBU", "Beijing Normal-Hong Kong Baptist University," Bnbu.edu.cn, 2018. https://www.bnbu.edu.cn/index.htm

[17] P. C. Simeao, "Historic canal houses and boats in Amsterdam at sunset - Netherlands,", "Stock photos, royalty-free images & video clips | iStock," iStockPhoto.com, 2025. https://www.istockphoto.com/

[18] “Apple Leaf Pattern”, “Photophoto Image Gallery – Free Design Material Download Website and Chinese Graphic Design Template Sharing Platform,” Photophoto.cn. https://www.photophoto.cn/ (accessed May 20, 2026).

[19] “Japanese-style commuting attire”, “Pinterest,” Pinterest. https://ph.pinterest.com/

[20] “Seamless pattern with random blue dots on white background”, “Stock photos, royalty-free images, graphics, vectors & videos,” Adobe Stock. https://stock.adobe.com/ca

[21] “Hand-drawn Cute Polka Dot Pattern”, 588ku.com, 2026. https://588ku.com/so/45492617/ (accessed May 20, 2026).

[22] “Stylized Handwritten Letter B Vector Icon”, 588ku.com, 2026. https://588ku.com/so/45492617/ (accessed May 20, 2026).

[23] “Elegant Floral Swirl and Butterfly Vector Background”, “Photophoto Image Gallery – Free Design Material Download Website and Chinese Graphic Design Template Sharing Platform,” Photophoto.cn. https://www.photophoto.cn/ (accessed May 20, 2026).